Fiscal Year:	FY 2017	**Task Last Updated:**	FY 02/27/2017
PI Name:	Quintiere, James Ph.D.		
Project Title:	Experimental Investigation of Emulated Burning Rate at Various Gravity Levels		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jimq@umd.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	301-405-3993
Organization Name:	University of Maryland		
PI Address 1:	Fire Protection Engineering		
PI Address 2:	3104G Jm Patterson		
PI Web Page:			
City:	College Park	**State:**	MD
Zip Code:	20742-3031	**Congressional District:**	5
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2009 Combustion Science NNH09ZTT001N
Start Date:	01/01/2015	**End Date:**	12/31/2019
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	2
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
Contact Email:	dennis.p.stocker@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: Extended to 12/31/2019 per NSSC information (Ed., 2/3/17)		
Key Personnel Changes/Previous PI:	February 2017 report: None.		
COI Name (Institution):	Sunderland, Peter Ph.D. (University of Maryland) deRis, John Ph.D. (FM Global)		
Grant/Contract No.:	NNX15AD06A		
Performance Goal No.:			
Performance Goal Text:			
	NOTE: Continuation of "Experimental Investigation of Emulated Burning Rate at Various Gravity Levels," grant NNX10AD98G, with the same PI (Ed., 1/14/2015) A novel experimental apparatus, the Burning Rate Emulator (BRE), is being developed. This apparatus is capable of assessing ignitability, steady burning rate, and extinction characteristics of 'virtual' solid and liquid fuels as a function of gravity level, flow characteristics, and orientation. By changing the fuel gas and its flow rate, it will be possible to emulate steady combustion for a range of liquid and solid fuels having B numbers (mass transfer numbers) and radiation characteristics that fall anywhere within physically realistic bounds. Measurements of the critical mass flux at ignition can also be made. These measurements will be conducted as a function of scale and orientation to assess turbulent and laminar burning in natural convection. The experiment is well suited to testing in the Combustion Integrated Rack aboard		

Task Description:	the International Space Station (ISS). The Burning Rate Emulator (BRE) will simulate the steady burning of liquids and solids using a rectangular porous burner fed by gaseous fuel and equipped with a total heat flux sensor. BRE will control the fuel flow rate based on the measured heat flux at the burner and the measured surface temperature. This will allow it to use a small number of gaseous fuels to match the following four properties of a large number of condensed fuels: heat of combustion (LHV), heat of vaporization (L), surface temperature (Ts), and laminar smoke point. Extinguishment will be considered via changes in fuel flow rate. Both low flow rates (for kinetic extinction) and high flow rates (for radiative extinction) will be considered. The BRE flight program seeks to establish the burning conditions in a quiescent microgravity ambient in terms of material properties. The specific flight objectives are as follows: 1. Observe burning behavior in long-duration microgravity for CH4/N2 and C2H4/N2 mixtures on porous circular burners of 25 and 50 mm diameter. Consider O2/N2 ambients with various pressures and ambient O2 concentrations and pressures. NASA-considered atmosphere for human space missions will be examined: 14.7 psia, 21% oxygen: 10.2 psia, 26.5%; and 8.2 psia, 34% oxygen. Emphasize fuel conditions that can be related to a diverse range of condensed fuels. 2. Consider the conditions of Objective 1 for ambients of O2/CO2 (desired). 3. Consider the conditions of Objectives 1 – 2 for extinction limits.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project is expected to improve the understanding of fire and combustion processes in both microgravity and normal gravity. It also could advance the state of the art in the design of porous burners and simple heat flux meters.
Task Progress:	One journal paper was published in 2016 and a second one submitted last year has just been published in January 2017. Three conference papers or abstracts were published and presented. Two BRE spaceflight burners were sent to UMD (University of Maryland) for flow and flame characterization and heat flux gauge calibration. This was for both transient and steady operation, and accounts for different paints on different surfaces. The burners were then returned to NASA Glenn Research Center (GRC). The spaceflight burners are sufficient to achieve the BRE flight objectives pending microgravity flame testing. The 5 s BRE drop test program at GRC has completed. A total of 51 drop tests were performed. The results are being scrutinized to determine accurate rates of heat flux to the burner surface. An improved calibration rig for heat flux meters was developed. Calibration of heat flux gauges was performed, particularly our standard meters traceable to NIST (National Institute of Standards and Technology) and MedTherm. A model was developed to estimate the incident heat flux at the burner surface using the copper plate as a calorimeter. This will provide an independent measurement of heat flux to the burner surface (in addition to the heat flux gauges). The model considers transient heating, radiative emission, convection to the surroundings, conduction into the burner walls, and conduction into the heat flux gauges. The model has been applied to the spaceflight BRE burners and the UMD burners. Normal gravity tests were conducted on the UMD burners in order to validate the calorimeter model. The model was used to predict the average heat flux for the drop-tower tests. We are able to predict the average heat flux for these tests. This alternative approach to calculate heat flux is compared to the theoretical heat flux predicted by the analytical model of H. Baum.
Bibliography Type:	Description: (Last Updated: 10/08/2025)
Abstracts for Journals and Proceedings	Quintiere JG, Markan A, Sunderland PB, deRis J, Stocker DP. "Development of Spaceflight BRE Burners." 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. , Oct-2016
Articles in Peer-reviewed Journals	Zhang Y, Kim M, Sunderland PB, Quintiere JG, de Ris J. "A burner to emulate condensed phase fuels." Experimental Thermal and Fluid Science. 2016 May;73:87-93. http://dx.doi.org/10.1016/j.expthermflusci.2015.09.025 , May-2016
Articles in Peer-reviewed Journals	Lundstrom FV, Sunderland PB, Quintiere JG, van Hees P, deRis JL. "Study of ignition and extinction of small-scale fires in experiments with an emulating gas burner." Fire Safety Journal. 2017 Jan;87:18-24. http://dx.doi.org/10.1016/j.firesaf.2016.11.003 , Jan-2017
Papers from Meeting Proceedings	Quintiere JG, Markan A, Sunderland PB, deRis J, Stocker DP. "Microgravity Burning Using a Gas Burner Emulator." Eastern States Section of the Combustion Institute Spring Meeting, Princeton, New Jersey, March 13-16, 2016. Eastern States Section of the Combustion Institute Spring Meeting, Princeton, New Jersey, March 13-16, 2016. 6 p. , Mar-2016